

PRACTICAL TEST AUTOMATION WITH PLAYWRIGHT

Test Data Management Template

Keep test data clean, isolated and maintainable.

Introduction

Poor test data causes flaky, order-dependent tests. This template helps you manage data so each test is independent and repeatable.

Principles

- Each test owns its data; no test depends on another.
- Create what you need (preferably via API), then clean up.
- Avoid relying on pre-existing shared records.

Approaches

Approach	When to use
Inline data	Simple, self-contained tests
Data files (JSON)	Reusable fixtures and parameterised tests
Factory functions	Generating varied valid records
API setup/teardown	Fast, reliable state for UI tests

Worked Example

A test generates a unique email (e.g. user+timestamp), creates the account via API, runs the UI behaviour, then deletes the account in teardown — so it can run any number of times in any order.

Blank Reusable Template

- Data needed:
- How it is created:
- How uniqueness is ensured:

- Cleanup/teardown:
- Independent of other tests? (Y/N)

Practical Exercise

Take a test that reuses shared data and refactor it to create its own unique data and clean up afterwards. Run it twice in a row to confirm independence.

Best Practices

- Generate unique data to avoid collisions.
- Prefer API setup/teardown.
- Never depend on test execution order.

Common Mistakes

- Tests sharing one record and interfering.
- Hard-coded data that breaks on re-run.
- No cleanup, leaving environments messy.

INSIDE STLC PRO TIP

A reliable suite has tests that pass in any order, run alone or together, any number of times. If yours do not, data coupling is almost always the cause.